

Abstract & Problem Definition

1. Abstract

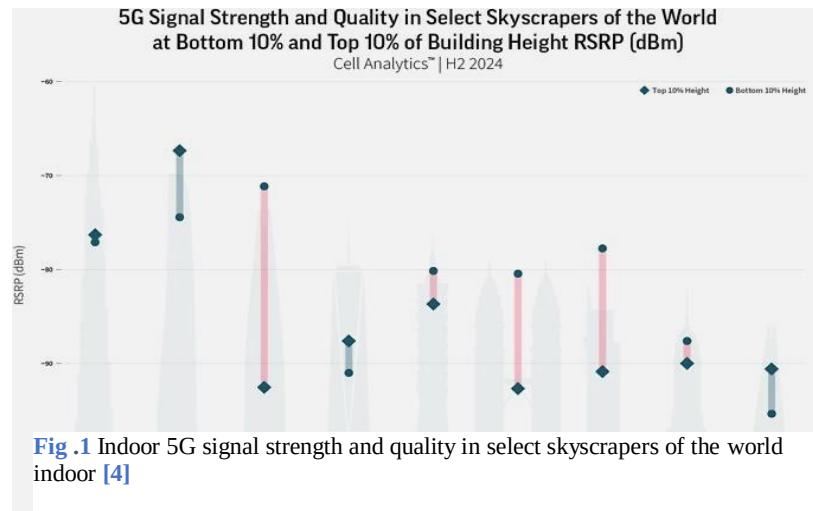
In an era of increasing data demand and **IoT** expansion, traditional static network distribution systems often suffer from signal attenuation, latency bottlenecks and uneven coverage. **Therefore**, in this project we'll explore the enhancement of network efficiency through the strategic placement and configuration of signal distribution nodes and complex phased antenna systems. The project also aims to achieve a balance between signal integrity and infrastructure costs. Our research is also supported with a hardware prototype (**PoC**).

2. Background and Motivation:

Over the last few decades, the importance of the internet has significantly increased. By **2025**, **74%** of the world population had internet signal coverage [1]. This accentuates the need for strong internet signal access everywhere around the world.

Connectivity problems affect people in various places. **68%** out of 4,000 American citizens had faced trouble with their internet connection in their homes last year. **76%** of them suffered from loss of signal in specific rooms “**dead zones**” [2]. Poor indoor internet coverage affects the economy as well, as it costs the UK economy as much as **100 billion** Pound Sterling annually, around **24 billion** of that is lost from the professional service sector. In the UK, **87%** of organizations announced that poor connectivity disrupts their work daily [3].

Height also plays a big factor in signal distribution, **Petronas Twin Towers** recorded **-80.44** dBm at ground levels, but it dropped to **-92.67** dBm at the top level. **The Lotte World Tower** recorded **-71.14** dBm at the lower floors, which dropped to **-92.54** dBm at the highest point, as shown in **Fig 1** [4]. These measurements are also affected by other factors, like the building materials of the tower and the network infrastructure. Some skyscrapers were able to work through all these problems, **Shanghai Tower** for example.



The in-building wireless market size was **22.88 billion** dollars in **2025**, and it's expected to become **66.15 billion** by **2035**, with an increasing rate of **11.2% annually** [5]. Which highlights the amount of efforts being exerted to ensure great connection everywhere.

3.Problem Statement:

In wide -range complex electrical communication systems, signal distribution is **rarely** uniform, and many resources are exhausted in overwhelmed regions while leaving others deficient, at such a large scale the challenge is keeping the system integrity while providing a balanced distribution.

Non-uniform internet signal distribution is mostly caused by the following problems : signal attenuation over long distances, causing null spots which are due to the destructive interference of waves. Overlapping between surrounding networks in the area resulting in noise. Physical obstacles like walls and buildings block the signals or cause reflections distorting them, this is especially noticeable in indoor locations.

Potential solutions for this problem:

- Using repeaters, which receive the signal, stop it , and retransmit a new strong clear one without any noise.
- Using low- frequency signals such as 4G instead of 5G, as they're much more durable over long distances and less sensitive to noise.
- Distributed antenna system (**DAS**) instead of using one tower, multiple small antennas are used in places where they're precisely needed.
- Creating multiple nodes which indicate where the dead zones are. The central **IC** adjusts the signal by taking advantage of constructive and destructive interference.

In this report we'll discuss and focus on the last solution.

4. Research objectives:

- Design and implement a system that uses **interference** with **antenna phased arrays** to readjust system distribution.
 - Design **a cost effective** node arrangement located at different places to ensure **full** area coverage.
 - Evaluate the system's performance through **simulations** and **hardware testing** to analyze its efficiency and results.
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